Collective Communication in AI Infra

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1 Introduction

Collective communication is defined as communication that involves a group of processes. These basic concepts are widely used in AI Infra:

- Broadcast (1:N): one member sends copies to all group members.
- Scatter (1:N): one member scatters data to all.
- Gather (1:N): one member gathers data from all.
- AllGather (N:N): each member gathers data from all, so every member ends up with the complete, identical dataset.
- AlltoAll (N:N): changes the data layout through dispatch + combine; the example here performs a matrix transpose.

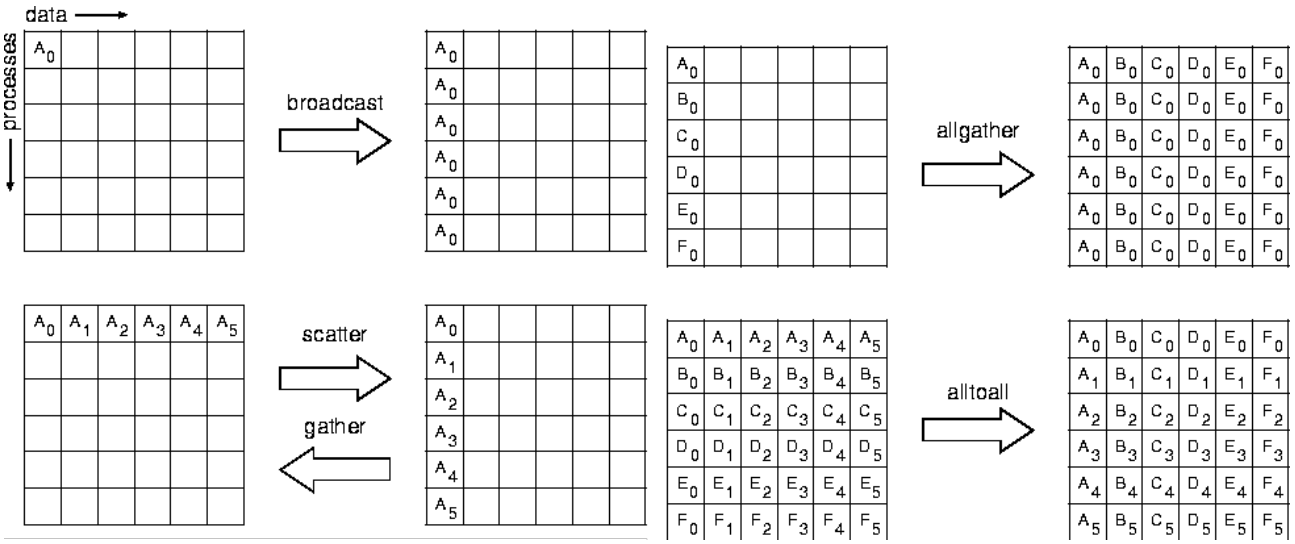


Figure 1: Collective communication patterns.

1.1 Reduce

Reduce is a special primitive: in terms of its result it is equivalent to Gather plus a Σ combine. However, to cut communication cost, implementations perform partial reduction, which is valid as long as the operation is **associative**. MapReduce sets an optional Combiner for exactly this reason [2].

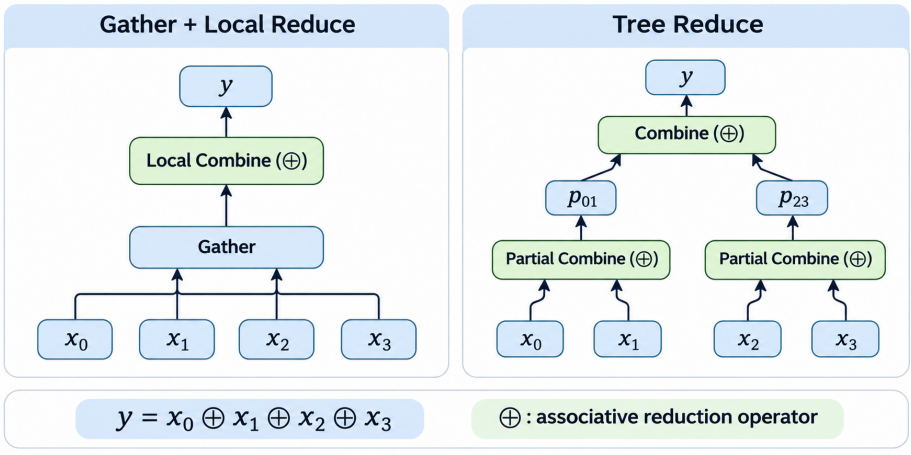


Figure 2: Two Reduce implementations.

2 AllReduce

AllReduce is a composite operation of:

1. **ReduceScatter**: distributes a global reduction task to nodes (GPU or server) by data shards.
2. **AllGather**: copies the finished shards to all nodes.

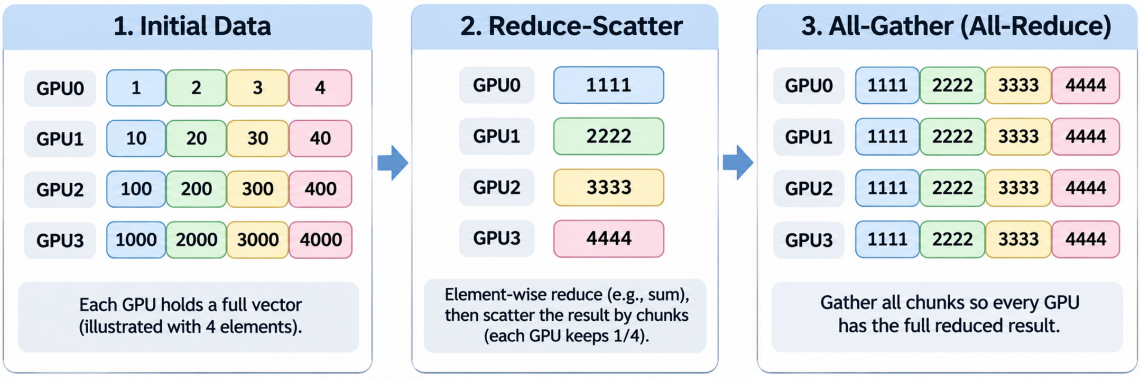


Figure 3: AllReduce.

2.1 Ring AllReduce

The Ring implementation is theoretically **bandwidth-optimal**: with N nodes and a per-node data size of S, it runs in $2(N-1)$ steps, each sending and receiving S/N data, so the per-node traffic(only send volume) is

$$2(N-1)\cdot\frac{S}{N}=2S\,\frac{N-1}{N}\xrightarrow{N\to\infty}2S,$$

which is asymptotically independent of the number of nodes.

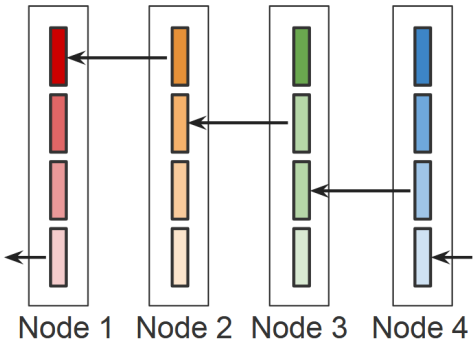


Figure 4: Ring AllReduce.

2.2 AllReduce in AI Infra

DDP (Distributed Data Parallel): training frequently uses AllReduce. Each device holds the complete parameters and a batch, and computes the local gradient g_i; each device is responsible for aggregating the shard g_j, and finally they exchange the shards to obtain the final gradient g.

FSDP (Fully Sharded Data Parallel): similar to DDP, but the parameters are also sharded, so the AllGather step is not needed.

TP (Tensor Parallel): For inference, TODO

References

[1] MPI Forum, *MPI: A Message-Passing Interface Standard*, Version 1.1, Section 4.1, "Introduction and Overview."

[2] Jeffrey Dean and Sanjay Ghemawat, *MapReduce: Simplified Data Processing on Large Clusters*, OSDI 2004.